

Fine-tune a Large Language Model for an Oracle APEX Application

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About Adrian

- Director of Innovation, AI and Cloud Solutions @ Insum
- Specialises in APEX and Oracle Cloud Infrastructure
- Master of Technology in Knowledge Engineering,
National University of Singapore
- Oracle certified professional
- Oracle ACE Director
- Book author
- Blogs at <https://fuzziebrain.com>



 Oracle ACE
Director



<https://www.apress.com/book/9781484260104>



<https://www.apress.com/book/9781484281697>



RAPID ORACLE SOFTWARE SOLUTIONS FOR FUTURE-READY BUSINESSES



21 Years in Business



36 Oracle APEX Certified Employees



58 OCI Certifications



14 Countries - Global Presence



11 Oracle ACEs/Oracle ACE Director



ISO 9001:2015 Certified



2023 Intel Oracle Global Leaders Partner Award



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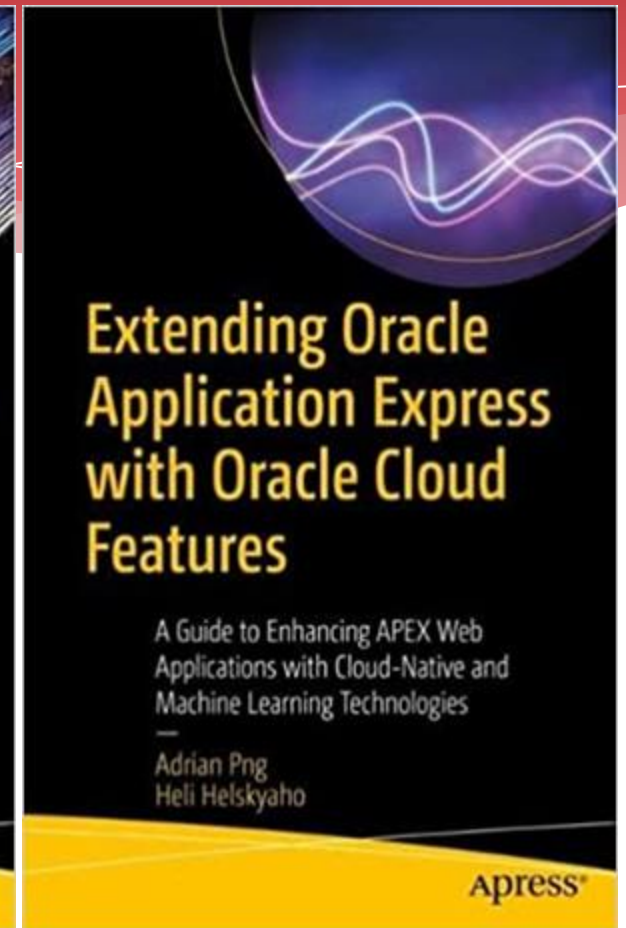
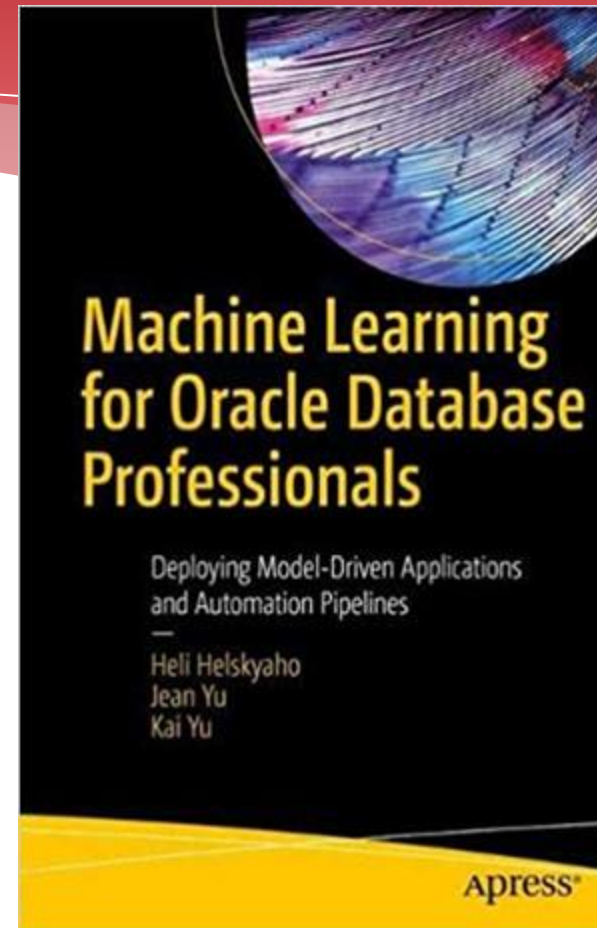
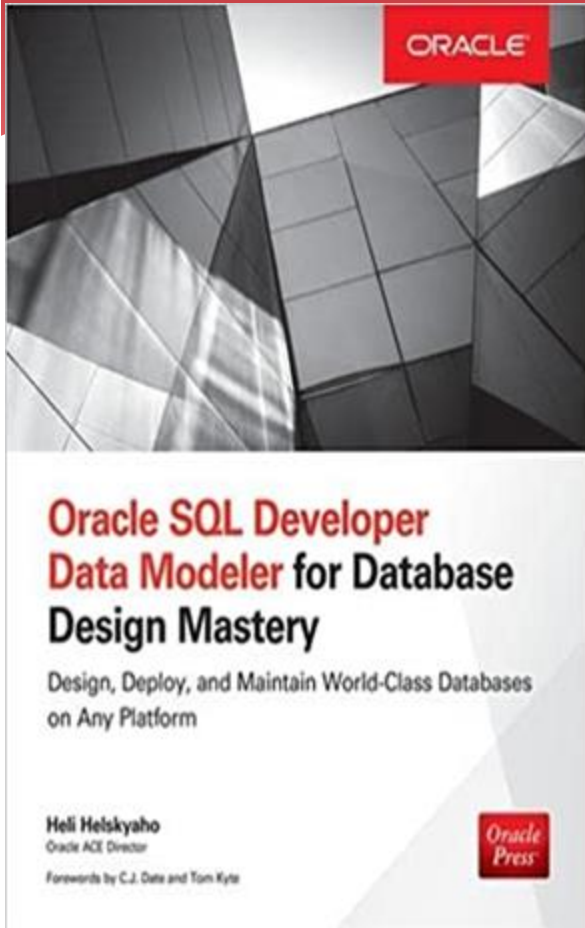
Heli



- * Graduated from University of Helsinki (Master of Science, computer science), currently a doctoral student, researcher and lecturer at University of Helsinki
- * Worked with Oracle products since 1993, worked for IT since 1990
- * Data and Database!
- * CEO for Miracle Finland Oy
- * Oracle ACE Director
- * Public speaker and an author
- * Author of the book Oracle SQL Developer Data Modeler for Database Design Mastery (Oracle Press, 2015), co-author for Real World SQL and PL/SQL: Advice from the Experts (Oracle Press, 2016), Machine Learning for Oracle Database Professionals: Deploying Model-Driven Applications and Automation Pipelines (Apress, 2021), and Extending Oracle Application Express with Oracle Cloud Features: A Guide to Enhancing APEX Web Applications with Cloud-Native and Machine Learning Technologies (Apress, 2022)



Books



Objective

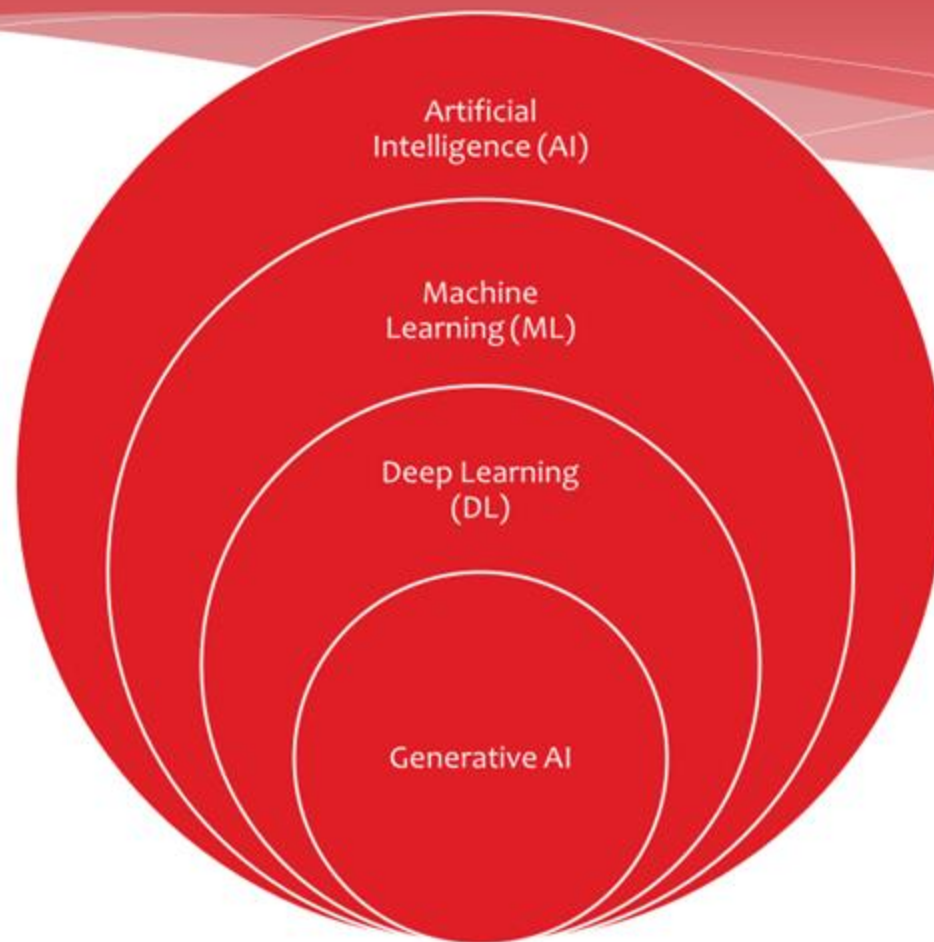
Learn about fine-tuning large language models using the OCI Data Science service, and then applying the use of these models in a web application.



Agenda

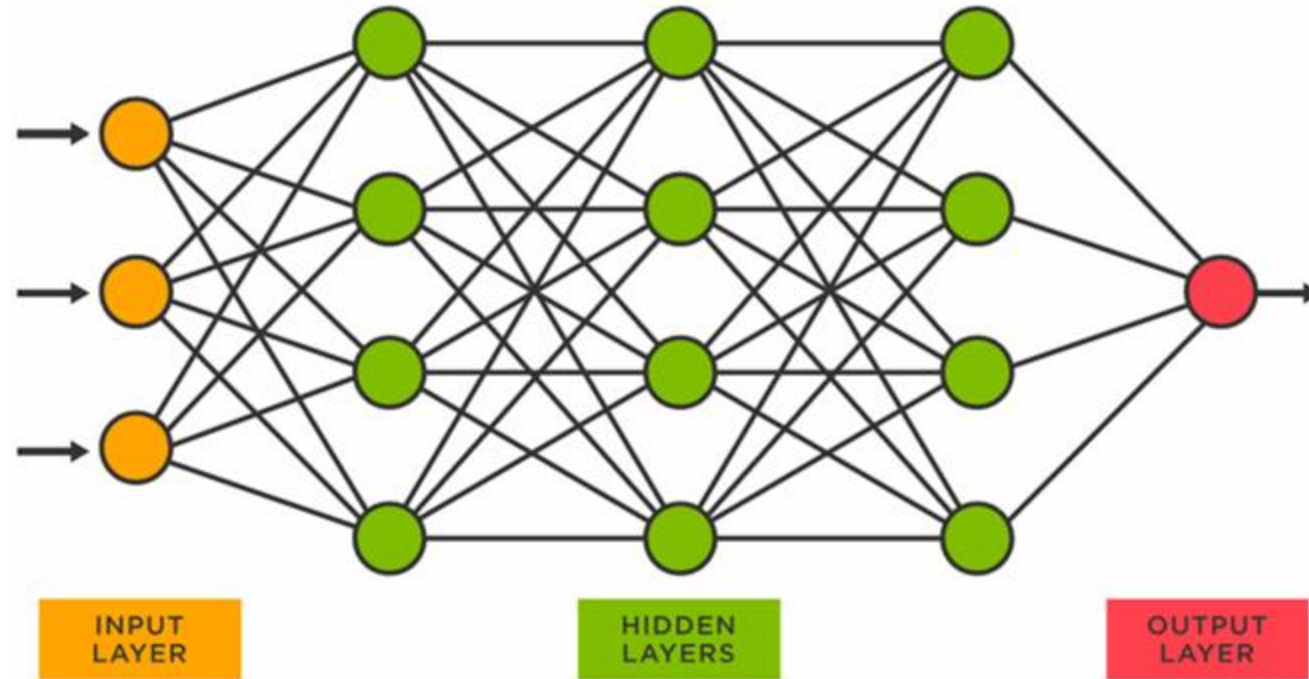
- Introduction to Transformers
- Why fine-tune an LLM?
- Why “go local”?
- Platform options and OCI Data Science service overview
- How to fine-tune the LLM and use it for inference tasks

Generative AI



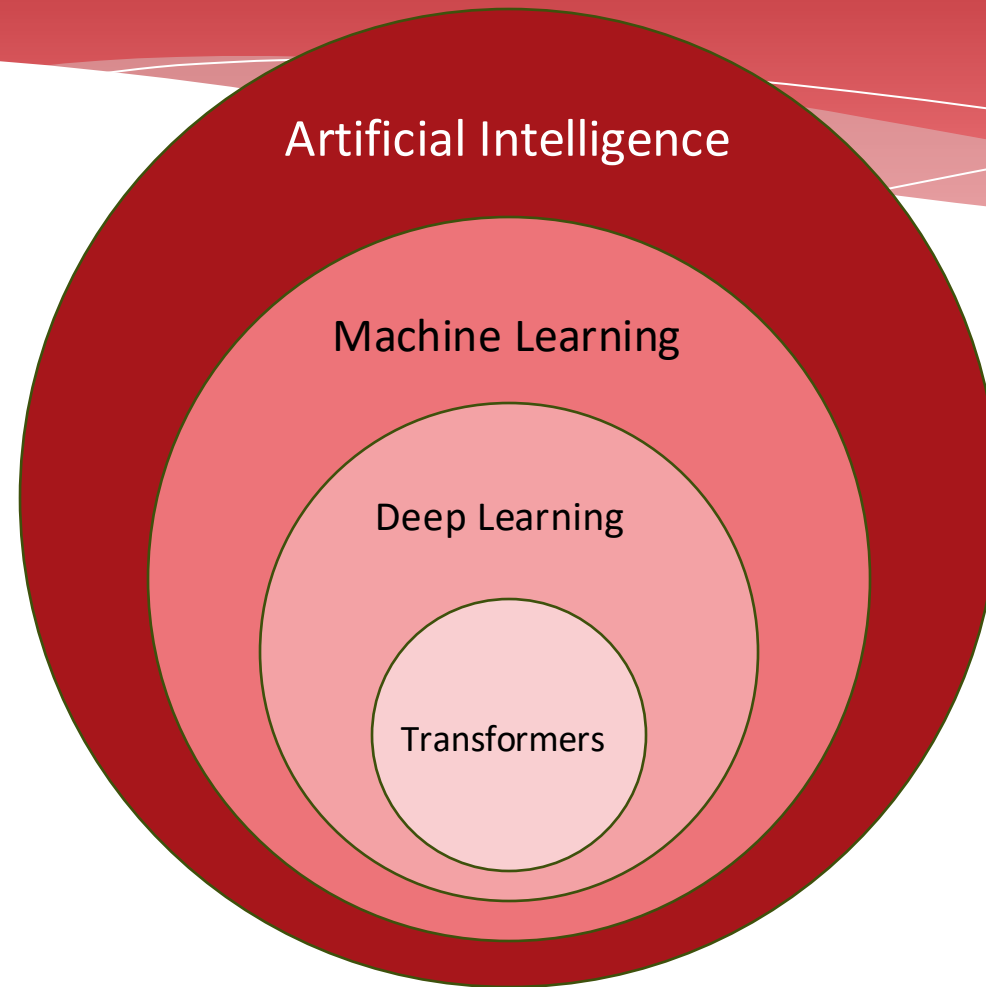
Deep Learning and Neural Networks (NN)

This
presentation
is



Word	Probability
Excellent	0.801
Cat	0.001
...	
Confusing	0.512

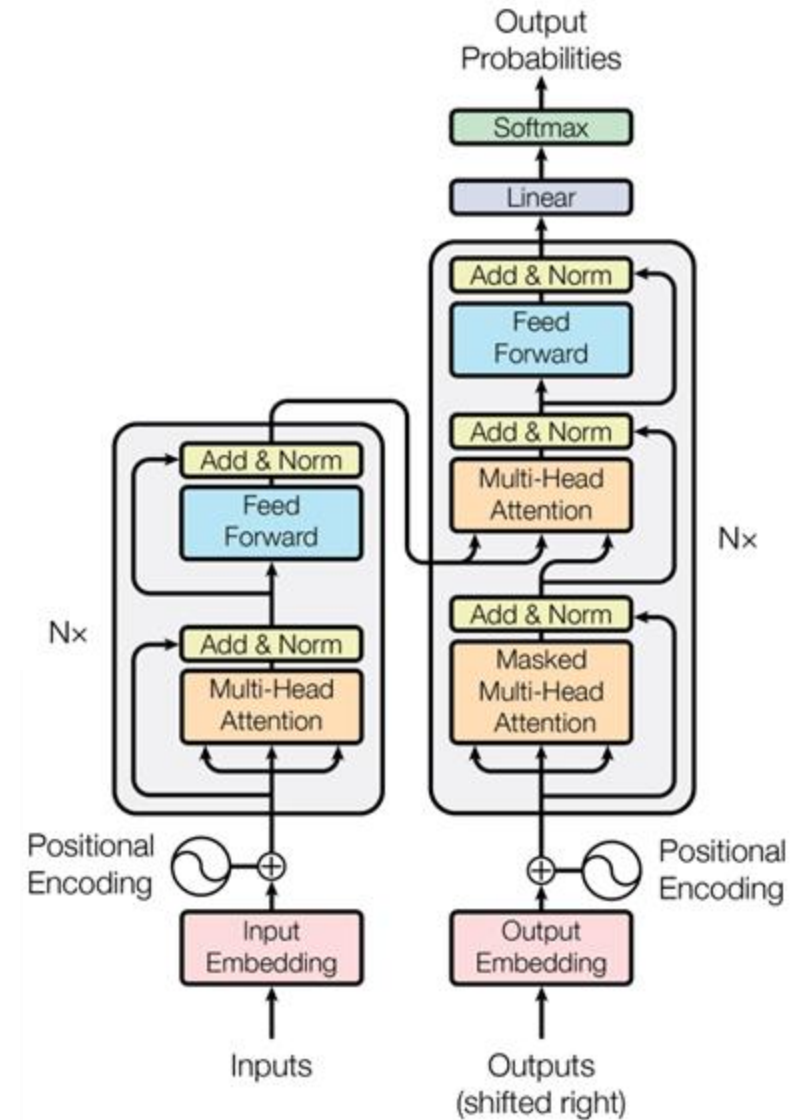
Transformers?



Pay Attention

- What are Transformers?

5.03762v7 [cs.CL] 2 Aug 2023

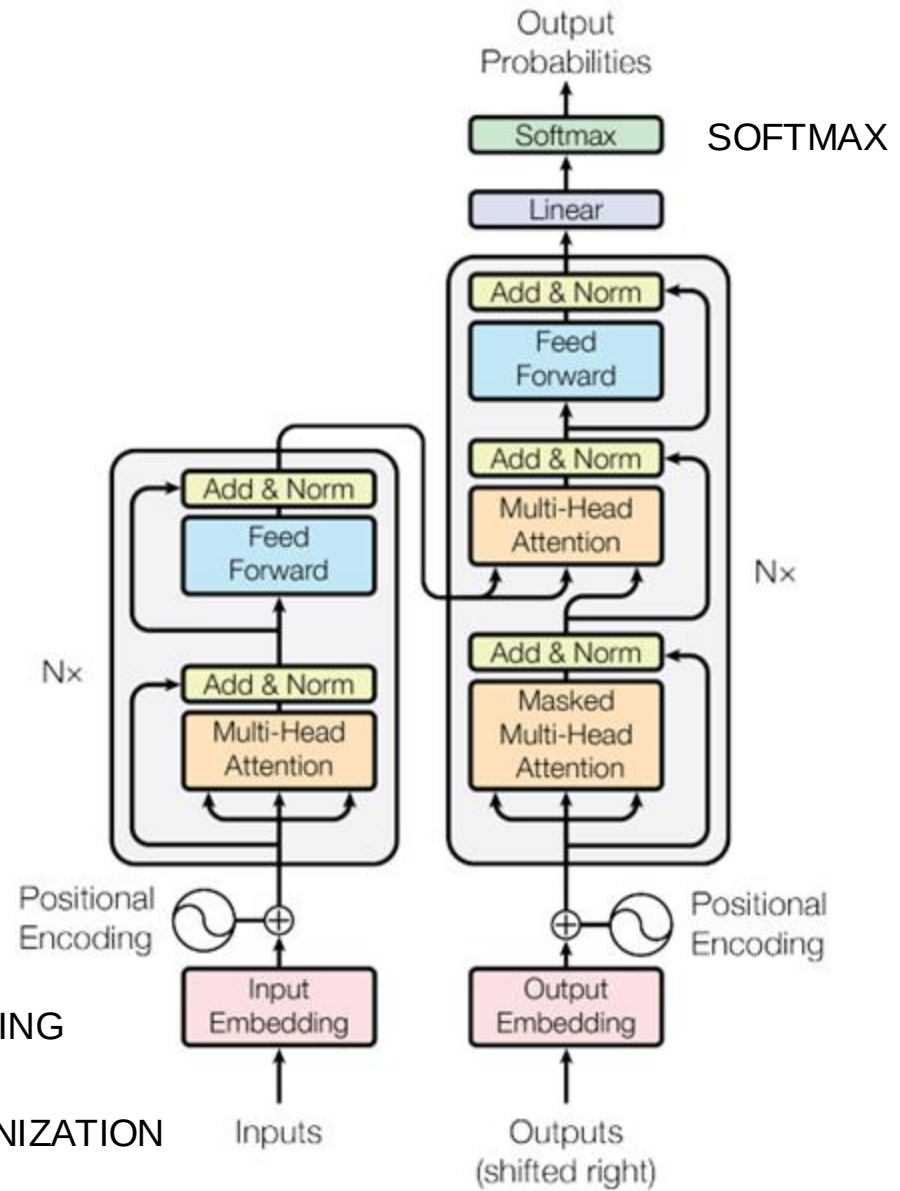


TRANSFORMER BLOCKS

POSITIONAL ENCODING

EMBEDDING

TOKENIZATION



Source: <https://doi.org/10.48550/arXiv.1706.03762>

TRANSFORMER EXPLAINER

Examples ▾

On the deserted planet they discovered a **mysterious**

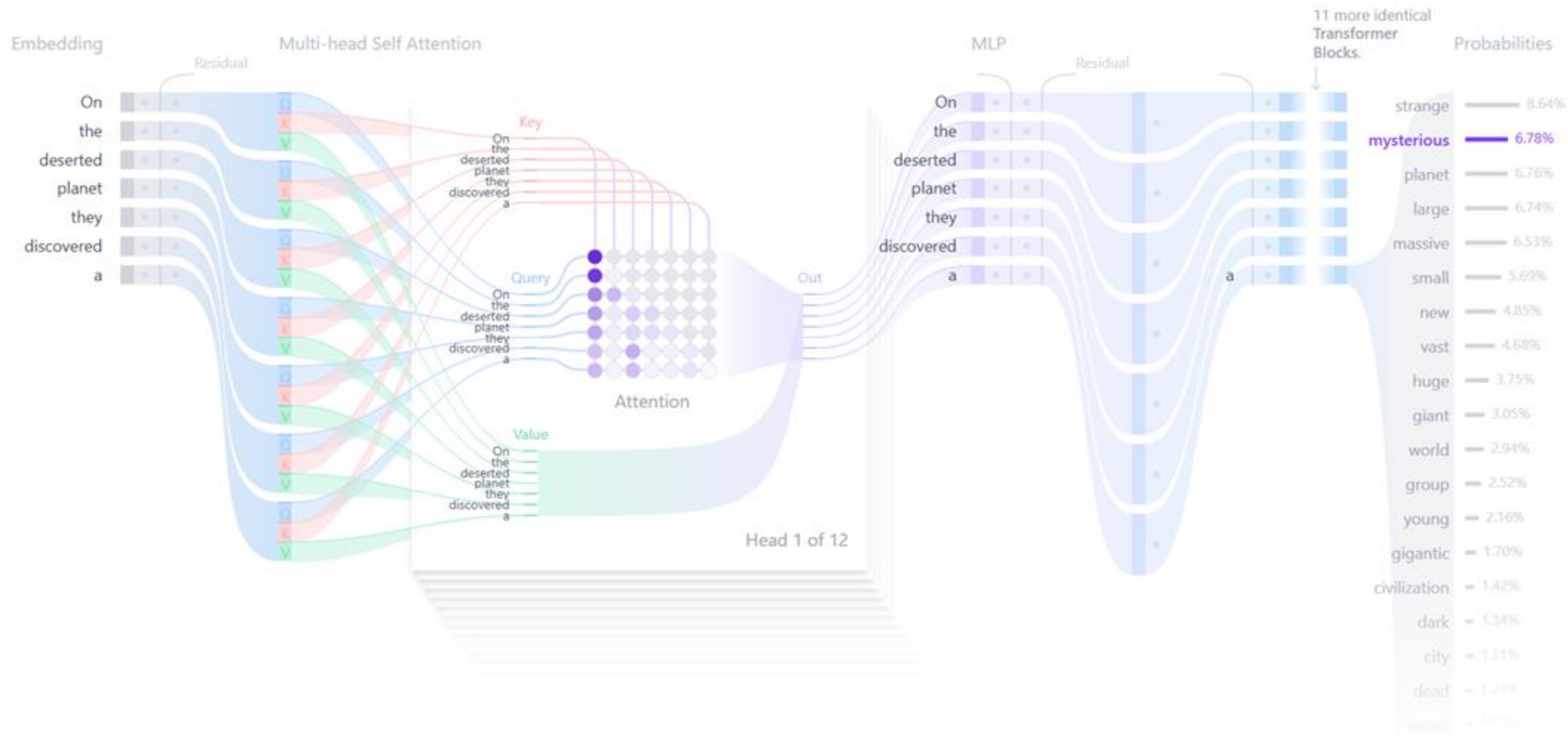
Generate

Temperature 1

1



Try the examples while GPT-2 model is being downloaded (600MB)





Large Language Models

- A foundation model that have been trained on large corpuses of textual information.
- Made popular by OpenAI with the launch of ChatGPT in late 2022.
- Many open-source models made available through [Hugging Face](#).

Why Fine-tune an LLM?



A pre-trained model may not be trained for specific domains and/or tasks.



Pre-training a model is time consuming and usually expensive.



Training/tweaking only a subset of model weights using a dataset of labelled data for the specific task or domain.



Environmental Impact

- Training a LLM from scratch requires not only extremely large datasets...
- They need vast amounts of:
 - Computational resources
 - Power
 - Time
 - Money
 - Knowledge
 - ...



WIRED

<https://www.wired.com> › Security › artificial intelligence

ChatGPT Has a Big Privacy Problem

Apr 4, 2023 — Europe's GDPR rules, which cover the way organizations collect, store, and use people's personal **data**, protect the **data** of more than 400 million ...



politico.eu

<https://www.politico.eu> › article › chatgpt-world-regulat...

ChatGPT is entering a world of regulatory pain in Europe

Apr 10, 2023 — France's **data protection** authority CNIL received at least two complaints against **ChatGPT**, on grounds of privacy violations including of the GDPR ...



PBS

<https://www.pbs.org> › newshour › economy › chatgpt-...

ChatGPT violated European Union privacy laws, Italy tells ...

Jan 30, 2024 — Italian regulators say they informed OpenAI that its **ChatGPT** artificial intelligence chatbot has violated **European Union's** stringent **data** ...



TechCrunch

<https://techcrunch.com> › chatgpt-italy-gdpr-notification

ChatGPT is violating Europe's privacy laws, Italian DPA ...

Jan 29, 2024 — OpenAI has been told it's suspected of violating **European Union privacy**, following a multi-month investigation of its AI chatbot, **ChatGPT**, ...



Forbes

<https://www.forbes.com> › Innovation › Cybersecurity

ChatGPT-4o Is Wildly Capable, But It Could Be A Privacy ...

May 17, 2024 — Yet **ChatGPT** has already made waves in **Europe** because its policies were previously contrary to the **EU General Data Protection Regulation (GDPR)**.

Keeping It “Local”

- For some organizations, keeping enterprise data within the boundaries is mandatory.
- Data residency, privacy, and confidentiality.

From theory to practice...

Training for Specific Tasks

- Publicly available LLMs are typically pretrained with publicly available data and for specific tasks like:
 - Text generation
 - Summarization
 - Text classification
- The model *bert-base-uncased* was pretrained to perform *masked language modeling* (MLM) that we will fine-tune to perform text classifications.

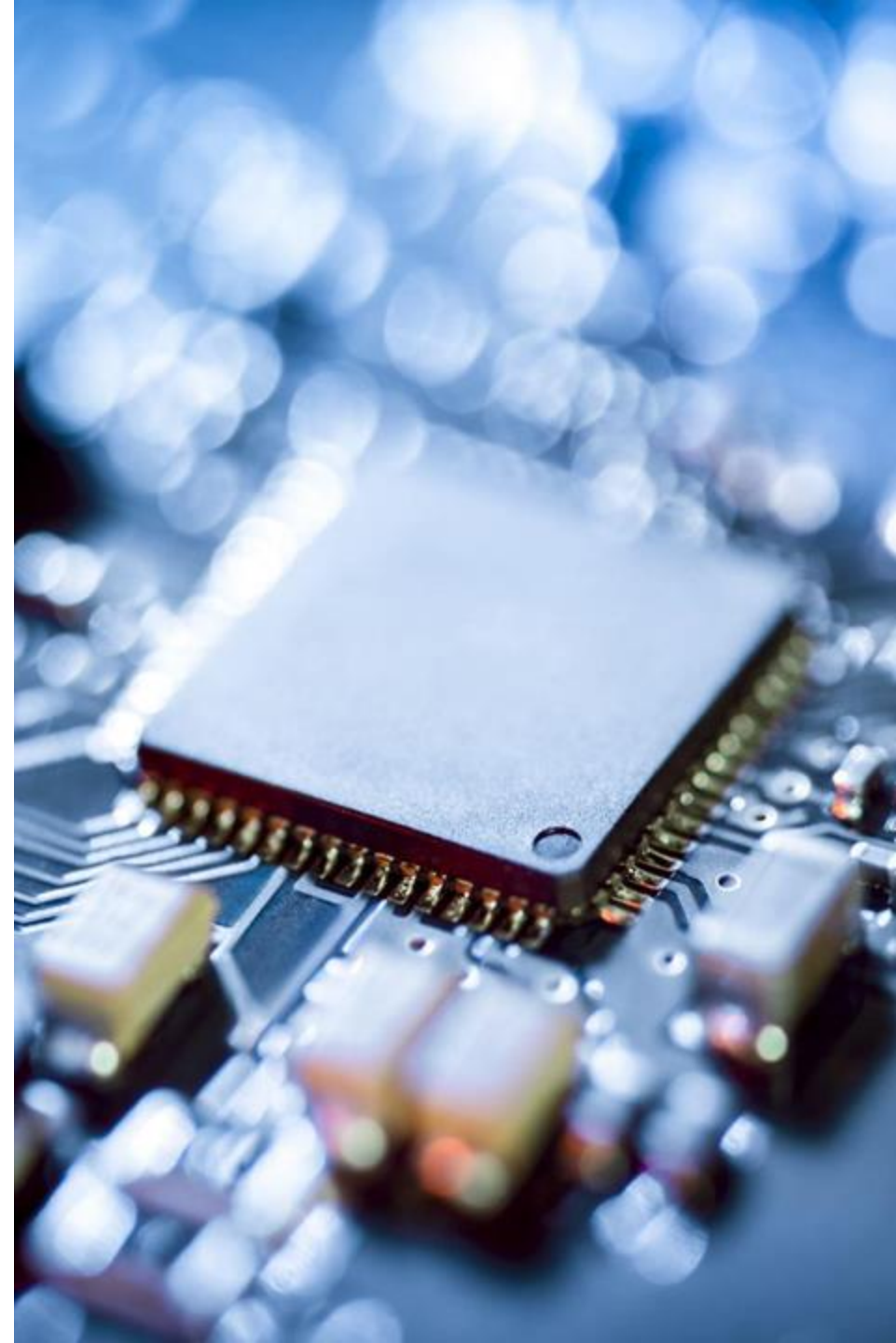
Intended uses & limitations

You can use the raw model for either masked language modeling or next sentence prediction, but it's mostly intended to be fine-tuned on a downstream task. See the [model hub](#) to look for fine-tuned versions of a task that interests you.

Note that this model is primarily aimed at being fine-tuned on tasks that use the whole sentence (potentially masked) to make decisions, such as sequence classification, token classification or question answering. For tasks such as text generation you should look at model like GPT2.

Platform Options

- OCI Generative AI's Dedicated AI Clusters
- OCI Data Science



Fine-Tuning on AI Dedicated Clusters

1. Create training dataset. Format in JSONL.
2. Upload to object storage.
3. Create a fine-tuning dedicated AI cluster.
4. Create a new custom model.
5. Create a hosting dedicated AI cluster.
6. Create an endpoint.
7. Consume the endpoint to perform inference task.



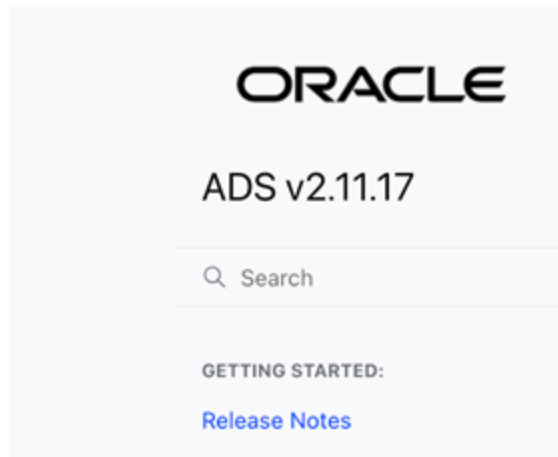
Challenges

- Service limits
- Minimum commit
- Restricted base foundation models

OCI Data Science Service

- A managed platform for building and deploying machine learning models quickly and easily.
- Features:
 - Notebook sessions (Jupyter)
 - Jobs
 - Pipelines
 - Model catalog
 - Model deployments





Oracle Accelerated Data Science (ADS)



 PYPI **V2.11.17**  PYTHON **3.8 | 3.9 | 3.10 | 3.11**  DOCS **NOTEBOOK-EXAMPLES**

Introducing AI Quick Actions

Deploy, Fine Tune and Evaluate Large language models such as [Mistral-7B-Instruct-v0.2](#), [CodeLlama-13b-Instruct-hf](#), etc with just a couple of clicks.

Oracle Accelerated Data Science (ADS)

- A Python package maintained by the OCI Data Science service team.
- Provides support for common data science tasks.
- Works well with OCI services.
 - For example, with OCI Data Science, manage lifecycle of ML model and perform deployments.

Projects

- Allows a logical separation of data science projects.
- Groups the resources notebook sessions, model catalog, model deployments etc.
- Use OCI policies to manage fine-grain access to projects.

Notebooks/Sessions

- JupyterLab, a web-based environment for working on code, data, and documentation.
- A notebook is a shareable document that allows authors to put write and describe code, create and display data visualisations, and more.
- OCI Data Science is a managed JupyterLab environment.
- Tailored for your computational needs:
 - Compute shape, includes GPU
 - Storage
 - Attach OCI file system storage
 - Set runtime environment variables
- Authenticate with IAM/IDCS
- Activate/deactivate



Jobs and Pipelines

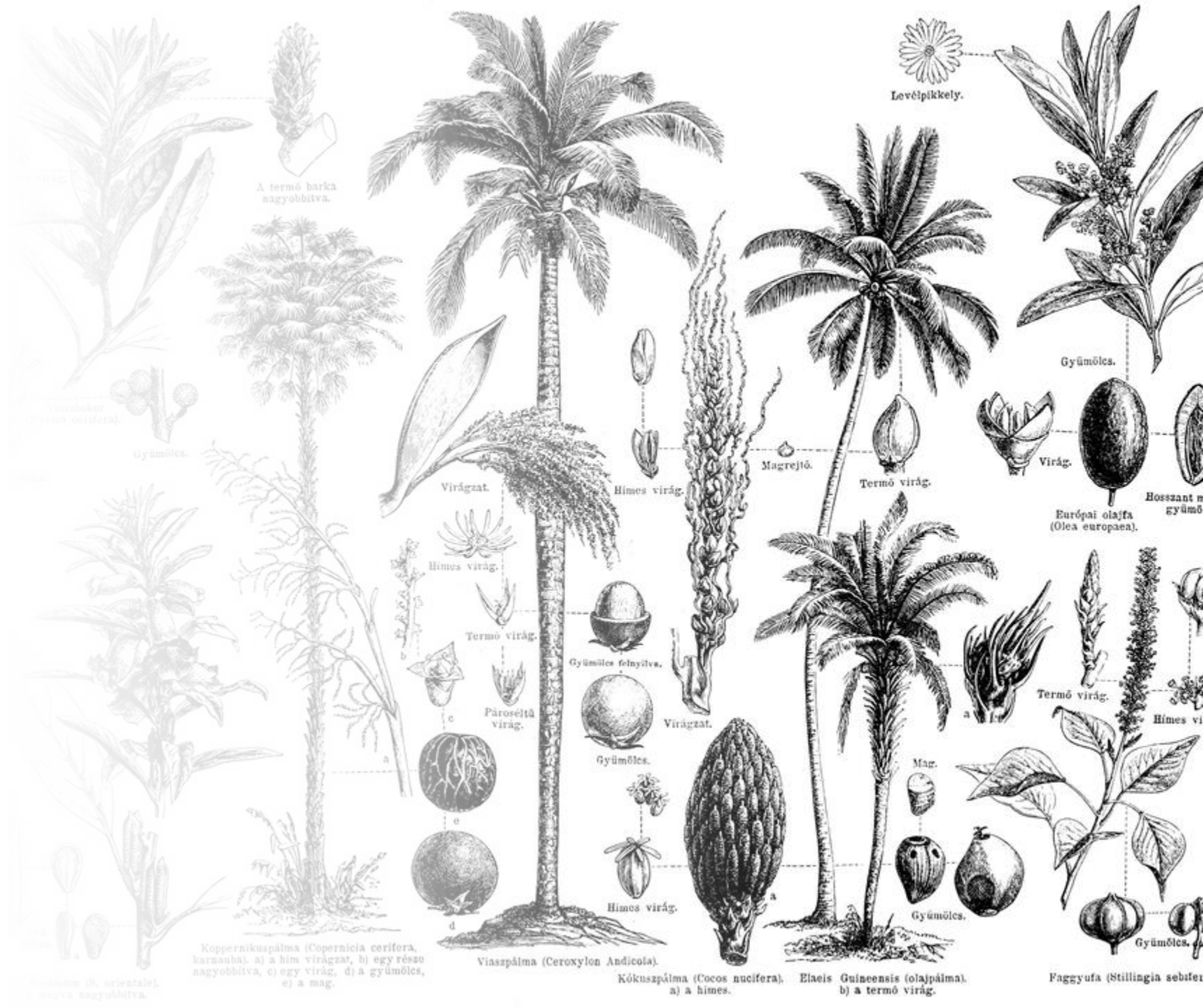


- Jobs
 - Run ML or data science tasks outside of notebook sessions.
 - Create reusable tasks.
 - Perform batch operations.

Pipelines

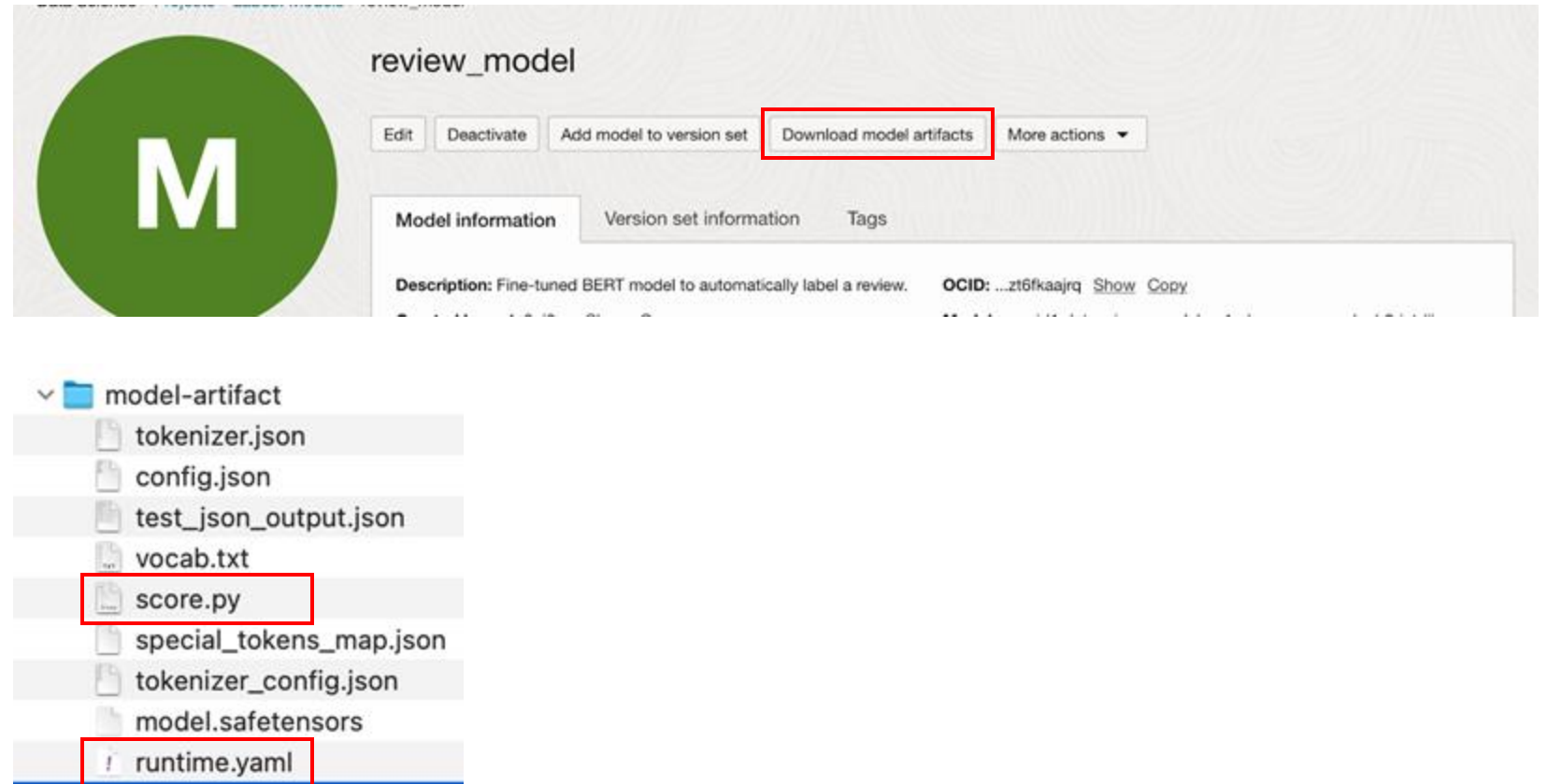
- Orchestrate a collection of tasks.
- Each task could run as a job, script, or within a containerized image.
- Often used to execute a ML workflow/pipeline.

-



Model Artifact

- Download
- Share
- Deploy



score.py

```
# score.py 1.0 generated by ADS 2.11.12
Import ...

task = 'sentiment-analysis'

@lru_cache(maxsize=10)
def load_model(model_file_name=""):
    ...
    return model

def predict(data, model=load_model()):
    ...
    inputs = pre_inference(data)

    yhat = post_inference(
        model(**inputs) if isinstance(inputs, dict) else model(inputs)
    )
    validated_model_outputs(yhat)
    return {'prediction': yhat}
```

runtime.yaml

```
MODEL_ARTIFACT_VERSION: '3.0'
MODEL_DEPLOYMENT:
  INFERENCE_CONDA_ENV:
    INFERENCE_ENV_PATH: oci://livelabs@***/conda_environments/gpu/review/1.0.0/review_v1_0_0
    INFERENCE_ENV_SLUG: review_v1_0_0
    INFERENCE_ENV_TYPE: published
    INFERENCE_PYTHON_VERSION: '3.9'
MODEL_PROVENANCE:
  PROJECT_OCID: ''
  TENANCY_OCID: ''
  TRAINING_CODE:
    ARTIFACT_DIRECTORY: /home/datascience/review_model_artifact
  TRAINING_COMPARTMENT_OCID: ''
  TRAINING_CONDA_ENV:
    TRAINING_ENV_PATH: oci://livelabs@***/conda_environments/gpu/review/1.0.0/review_v1_0_0
    TRAINING_ENV_SLUG: review_v1_0_0
    TRAINING_ENV_TYPE: published
    TRAINING_PYTHON_VERSION: '3.9'
  TRAINING_REGION: ''
  TRAINING_RESOURCE_OCID: ''
  USER_OCID: ''
  VM_IMAGE_INTERNAL_ID: ''
```

Model Taxonomy

- Model algorithm
- Use case
- Model framework
- Model hyperparameters

Taxonomy type	Value
Model algorithm/ model estimator object	TextClassificationPipeline
Use case	Sentiment analysis
Model Framework Version	4.41.2
Model hyperparameters	<pre>{ "return_dict": "True", "output_hidden_states": "False", "output_attentions": "False", "torch_dtype": "float32", "use_bfloat16": "False", "tf_legacy_loss": "False", "torchscript": "False", "adapters": null, "tie_word_embeddings": "True", "chunk_size_feed_forward": 0, "decoder": null, "_decoder": "False", "is_decoder": "False", "cross_attention_hidden_size": "None", "attention": null, "_attention": "False", "tie_encoder_decoder": "False", "max_length": "20", "min_length": "0", "do_sample": "False", "early_stopping": "False", "num_beams": "1", "num_beam_groups": "1", "diversity_penalty": "0.0", "temperature": "1.0", "top_k": "50", "top_p": "1.0", "typical_p": "1.0", "repetition_penalty": "1.0", "length_penalty": "1.0", "no_repeat_ngram_size": "0", "encoder_no_repeat_ngram_size": "0", "bad_words_ids": "None", "num_return_sequences": "1", "encoder_no_return_sequences": "False", "scores": "False", "return_dict_in_generate": "False", "forced_bos_token_id": "None", "forced_eos_token_id": "None", "remove_invalid_values": "False", "exponential_decay_length_penalty": "None", "suppress_tokens": "None", "begin_suppress_tokens": "None", "generation_config": { "BertForSequenceClassification": { "finetuning_task": "None", "id2label": { "0": "LABEL_1", "1": "LABEL_2", "2": "LABEL_3", "3": "LABEL_4" }, "label2id": { "LABEL_1": "0", "LABEL_2": "1", "LABEL_3": "2", "LABEL_4": "3" }, "tokenizer": null, "prefix": "None", "bos_token_id": "None", "pad_token_id": "0", "eos_token_id": "None", "decoder_start_token_id": "None", "task_specific_params": { "single_label_classification": { "_name_or_path": "./review_model", "revision": "4.41.2", "gradient_checkpointing": "False", "model_type": "bert", "vocab_size": "30024", "hidden_size": "768", "num_hidden_layers": "12", "num_attention_heads": "12", "intermediate_size": "3072", "hidden_dropout_prob": "0.1", "attention_probs_dropout_prob": "0.1", "max_position_embeddings": "512", "type_vocab_size": "2", "initializer_range": "0.02", "layer_norm_eps": "1e-12", "position_embedding_type": "absolute", "classifier_dropout": "None" } } } } }</pre>
Model Framework	Transformers

A photograph of a person lying in a hospital bed, wearing a light blue hospital gown. Their hands are clasped together on their lap. The background is softly blurred, showing a teal blanket and a white pillow. The overall tone is calm and clinical.

Model Deployments

- Uses managed OCI resources: compute and load balancers.
- No fuss deployment of the ML model as REST APIs.
- Auto-scaling easily configured.

How to fine-tune the LLM & use it for inference tasks

1. Prepare the OCI environment.
2. Perform the fine-tuning of a BERT model to assign a review score (sentiment) using the OCI Data Science service.
3. Deploy the LLM.
4. Create an Oracle APEX application that uses the fine-tuned LLM to predict the score for a new review entry.

Prepare the OCI Environment

- IAM resources:
 - Compartment
 - Group and user (for application “agent”), and generate API key
 - Dynamic group
 - Policies
- Oracle Autonomous Database and Oracle APEX workspace
- OCI Data Science project and notebook session

The Yelp Dataset for Training



- Yelp.com publishes reviews about business worldwide through user contributions.
- The company first made the dataset available to the public the Yelp Dataset Challenge in 2015 and continues to maintain and publish updates.
- It currently has almost 7 million reviews collected for more than 150K businesses.
- More information about this dataset can be found at:
<https://www.yelp.com/dataset>

Fine-tune the Base LLM

- Launch the notebook session and setup the runtime environment
- Import the required Python packages
- Load and prepare the Yelp review dataset
- Configure the trainer
- Train and test the model
- Save the model to the file system

<https://www.youtube.com/watch?v=lbAxLtlFr-A>



Deploy the Fine-tuned LLM

- Prepare the OCI environment and configure the notebook session to publish the Conda environments
- Publish the Conda environment
- Prepare the model artifact
- Register the model to the Model Catalog
- Deploy the model

<https://www.youtube.com/watch?v=lbAxLtIFr-A&t=124>



Use the Fine-tuned LLM in an Oracle APEX App

- Setup a workspace credential using the OCI API keys
- Create an application with a page to create new reviews
- Add a dynamic action in the data entry page to call the deployed model's "predict" endpoint
- Suggest a review rating using the predicted value

<https://www.youtube.com/watch?v=IbAxLtIFr-A&t=304>



The Workshop on Oracle LiveLabs



Go to <https://apexapps.oracle.com/pls/apex/r/dbpm/livelabs>, then search “3965”.

OR

Link: <https://apexapps.oracle.com/pls/apex/r/dbpm/livelabs/view-workshop?wid=3965>

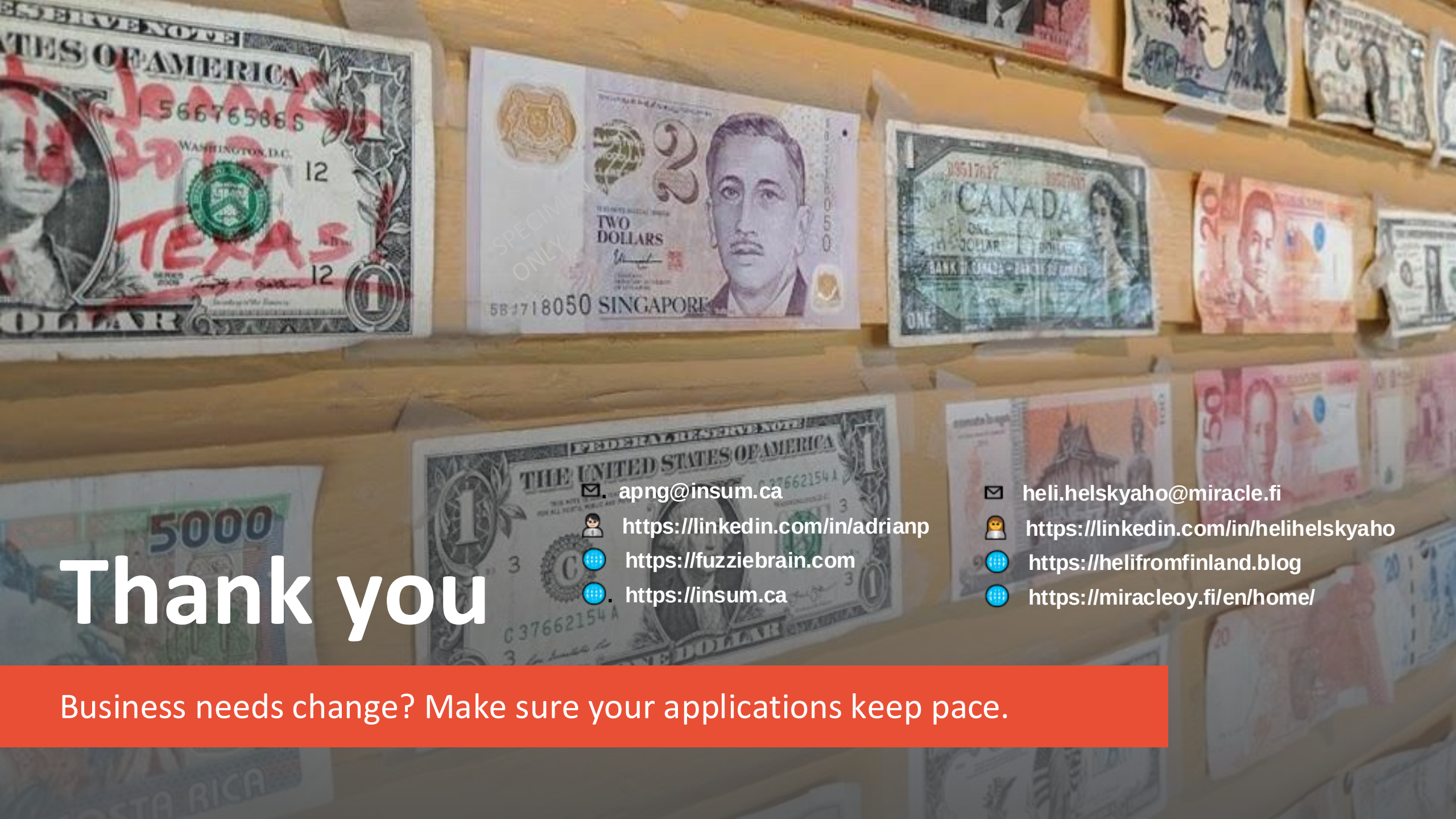


What Do You Need?

- An **OCI tenancy** and necessary privileges to create resources.
- Some knowledge about **machine learning**, **Python programming**, and **Oracle APEX**.
- Curiosity and passion to learn new things.



Questions?



Thank you

Business needs change? Make sure your applications keep pace.

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